Darcy First Proof

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1 Problem

Let $D \subseteq \mathbb{R}^d$, $d \geqslant 2$, denote the d-dimensional Euclidean unit ball. Given a function $f \in C^\alpha(D)$ with Hölder-Zygmund smoothness $\alpha > 1$ on D and let $u = u_f$ denote the unique solution of the following divergence form PDE (Darcy's problem)

$$\operatorname{div}(f\nabla u) = 1 \ \text{ on } D, \quad u|_{\partial D} = 0.$$

Consider the non-linear statistical inverse problem of reconstructing f from noisy measurements of u corrupted by Gaussian white noise

$$Y_n = u_f + \frac{1}{\sqrt{n}}\mathbb{W}.$$

For $r \in (0,1)$ consider the following class of conductivities

$$f \in \mathcal{F}(\alpha, r) = \{h \in C^\alpha(D),\ \|h - \mathbf{1}\|_{C^\alpha(D)} \leqslant r\},$$

where $\mathbf{1}$ denotes a constant function with value 1. For small (but fixed) $r \in (0,1)$, what is the minimax rate as $n \to \infty$ for estimating f in the L^2-mean squared error

$$\inf_{\hat{f}_N} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\left[\|\hat{f}_n - f_0\|^2_{L^2(D)}\right],$$

up to logarithmic factors?

1.1 Context

The Darcy problem is a prototypical non-linear statistical inverse problem, see e.g. [11, 6, 2], and is also a key example used for MCMC methodology in infinite-dimensions, e.g. [3, 5]. Minimax rates are currently unknown in all dimensions $d \geqslant 1$. Statistical consistency results with sub-optimal rates have been derived e.g. in [7, 10] and other references, where the best known rate is currently $n^{-\frac{\alpha-1}{2\alpha+2+d}}$, corresponding intuitively to a 2-ill posed rate with $(\alpha-1)$-smooth ground truth. The difficulty of the Darcy problem lies in that the problem has a 'mixed ill-posedness' which is spatially inhomogeneous. The location of 2-ill-posedness needs to be estimated in a data-driven way since it depends on the unknown f, making the current problem non-standard from a nonparametric statistical perspective.

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2 Solution

The main theorem states that the minimax rate corresponds to the rate expected in a 2-ill-posed inverse problem in $d-1$ dimensions with α-smooth ground truth.

2.1 Theorem. *Let $d \geqslant 2$ and $\alpha > 1$. There exists $r_0 > 0$ such that for all $r \leqslant r_0$, the minimax rate of estimation over $f \in \mathcal{F}(\alpha, r)$ in $L^2(D)$-loss is given, up to logarithmic factors, by $n^{-\alpha/(2\alpha+3+d)}$.*

The majority of the work consists of proving the upper bound. For any $x \in D$, we have the inversion formula

$$f(x) = \int_{-\infty}^{0} \exp\Big(- \int_s^0 \Delta u_f(\varphi(v))dv\Big)ds,$$

where φ is a characteristic given by the 'backward in time' ODE $\varphi'(s) = \nabla u_f(\varphi(s))$, $s \leqslant 0$ with initial condition $\varphi(0) = x$, see for example [9]. Note that φ depends on x through its initial condition.

Motivated by this, we study the following estimator. Let $K(\bullet)$ be a smooth kernel function with sufficiently many vanishing moments, satisfying $\int_{\mathbb{R}^d} K(y)dy = 1$ and support $\mathrm{supp}(K)$ in the unit ball D. For $K_\delta(y) = \delta^{-d}K(y/\delta)$, where the small bandwidth $\delta > 0$ is chosen later, we estimate

$$\hat{u}(y) = (K_\delta * Y_n)(y), \qquad \Delta\hat{u}(y) = \Delta(K_\delta * Y_n)(y) = \Delta K_\delta * Y_n(y).$$

Near the boundary of D, we employ a standard boundary correction for the kernel K (such as detailed in [1], which still has vanishing moments). We then consider for each $x \in D$ the estimator

$$\hat{f}(x) = \int_{-S}^{0} \exp\Big(- \int_s^0 \Delta\hat{u}(\bar{\varphi}(v))dv\Big)ds.$$

Throughout we will choose $S = C\log n$ for $C > 0$ being a sufficiently large constant, and $\bar{\varphi}(v)$ are 'pre-estimated' characteristics which we detail now.

2.1 Properties of the pre-estimator

We assume that $\bar{\varphi}$ arises from the gradient flow of an independent and sufficiently smooth pilot estimate $\bar{u} \in C^1$ of u_f. In particular we assume that such an estimator $\bar{u}$ is obtained via an independent sample (we can assume such sample splitting w.l.o.g., while increasing the noise level by a factor of $\sqrt{2}$) and that $\bar{\varphi}'(v) = \nabla\bar{u}(\bar{\varphi}(v))$, $v \leqslant 0$, with $\bar{\varphi}(0) = x$.

Since $f \in \mathcal{F}(r,\alpha) \subseteq C^\alpha(D)$, we have that $u_f \in C^{\alpha+1}(D)$ (see for instance [7]) whence we may assume that the pilot estimator $\bar{u}$ satisfies convergence rates

$$\mathbb{E}_f \|\bar{u} - u_f\|^2_{C^1} \leqslant C\big(\frac{\log n}{n}\big)^{\frac{2\alpha}{2\alpha+2+d}}, \ \mathbb{E}_f \|\bar{u} - u_f\|^2_{C^2} \leqslant C\big(\frac{\log n}{n}\big)^{\frac{2(\alpha-1)}{2(\alpha-1)+4+d}}, \tag{2.1}$$

for some $C > 0$. For $f = \mathbf{1}$ we have the explicit radial solution $u_{\mathbf{1}} = \frac{1}{2d}(\|x\|^2 - 1)$ which has constant Hessian $\nabla^2 u_{\mathbf{1}}(x) = \mathrm{diag}(1/d \dots 1/d)$. By Lemma 2.5 we can choose $r > 0$ small enough such that $\nabla^2 u_f(x)$ has minimal and maximal eigenvalues bounded below and above by $1/d - \eta/2$ and $1/d + \eta/2$ respectively, uniformly in f and x. Standard estimators for u_f, such as kernel-type or wavelet projection estimators, satisfy the above convergence rates as well as exponential concentration inequalities. In particular, one may assume that for any $\eta > 0$ there are constants $c_1, c_2, \gamma > 0$ (uniformly in $f \in \mathcal{F}(\alpha, r)$) such that

$$\mathbb{P}_f\big(\|\bar{u} - u_f\|_{C^2} \geqslant \eta/2\big) \leqslant c_1 \exp\big(-c_2 n^\gamma\big).$$

We may thus modify the pre-estimator $\bar{u}$ on an event (measurable with respect to the data) of probability at most $c_1 \exp\big(-c_2 n^\gamma\big)$ to ensure that with probability 1,

$$\forall x \in D: \ 1/d - \eta \leqslant \lambda_{min}[\nabla^2\bar{u}(x)] \leqslant \lambda_{max}[\nabla^2\bar{u}(x)] \leqslant 1/d + \eta,$$

while simultaneously satisfying the convergence rates (2.1).

In the next subsections, unless otherwise noted, expected values are taken only with respect to the statistical sample which supports the main estimator $\hat{u}$, and $\bar{u}$ may be treated as fixed and satisfying the above convergence rates. Throughout, $C^\alpha(D)$ denotes the α-Hölder-Zygmund smooth functions on D.

2.2 Bias

2.2 Proposition (Bias)**.** *Suppose that the bandwidth is chosen at least as large as* $\delta \gtrsim n^{-\frac{1}{2\alpha+d+3}}$. *Then for any* $x \in D$, *the bias is of order at most*

$$\mathbb{E}_f[\hat{f}(x)] - f(x) = O\big(\log(n)\delta^\alpha\big) + O\big(\log(n)\|\bar{u} - u_f\|_{C^1}\big).$$

Here, the constants implicit in $O(\bullet)$ *are uniform over* $x \in D$ *and* $f \in \mathcal{F}(\alpha, r)$.

Proof. Fix f. For simplicity we write $\mathbb{E}$ for $\mathbb{E}_f$, and u for u_f.
Step 1. The expectation of $\hat{f}(x)$ (conditional on $\bar{\varphi}$) can be written as

$$\begin{aligned}\mathbb{E}[\hat{f}(x)] &= \mathbb{E}\int_{-S}^0 \exp\Big(-\int_s^0 \Delta\hat{u}(\bar{\varphi}(v))dv\Big)ds \\ &= \int_{-S}^0 \exp\Big(-\int_s^0 [\Delta u * K_\delta](\bar{\varphi}(v))dv\Big) \times \mathbb{E}\Big[\exp\Big(-\int_s^0 \frac{1}{\sqrt{n}}[\mathbb{W} * \Delta K_\delta](\bar{\varphi}(v))dv\Big)\Big]ds.\end{aligned}$$

The expression inside the last exponential is centered normal with variance

$$\begin{aligned}\frac{1}{n}\,\mathbb{E}\Big[\Big(\int_s^0 [\mathbb{W} * \Delta K_\delta](\bar{\varphi}(v))dv\Big)^2\Big] &= \frac{1}{n}\int_{\mathbb{R}^d}\Big(\int_s^0 \Delta K_\delta(\bar{\varphi}(v) - z)dv\Big)^2 dz \\ &\leqslant \frac{|s|}{n}\int_{\mathbb{R}^d}\int_s^0 \Delta K_\delta(\bar{\varphi}(v) - z)^2 dv dz \\ &\lesssim \frac{s^2}{n}\delta^{-d-4},\end{aligned}$$

where $\|\Delta K_\delta\|_\infty \lesssim \delta^{-d-2}$ and the support of ΔK_δ in a ball of radius δ were used. By the exponential moment $\mathbb{E}[e^Z] = e^{\sigma^2/2}$ for $Z \sim N(0, \sigma^2)$ and $s^2 \lesssim \log(n)^2$ we thus find with a constant $C > 0$

$$\begin{aligned}\mathbb{E}\Big[\exp\Big(-\int_s^0 \frac{1}{\sqrt{n}}\mathbb{W} * \Delta K_\delta(\bar{\varphi}(v))dv\Big)\Big] &\leqslant \exp\Big(C\log(n)^2 n^{-1}\delta^{-(d+4)}\Big) \\ &= 1 + O\Big(\log(n)^2 n^{-1}\delta^{-(d+4)}\Big) \\ &= 1 + O(\delta^\alpha),\end{aligned}$$

where we used $n^{-1}\delta^{-(\alpha+d+4)} \lesssim n^{-(\alpha-1)/(2\alpha+d+3)} \lesssim (\log n)^{-2}$ by our choice of δ and $\alpha > 1$.

Step 2. Using $\Delta u = (1 - \nabla u_\bullet \nabla f)/f$, we have

$$\exp\Big(-\int_s^0 \Delta u * K_\delta(\bar{\varphi}(v))dv\Big) = \exp\Big(-\int_s^0 \frac{1}{f} * K_\delta\ (\bar{\varphi}(v))dv\Big)\exp\Big(\int_s^0 \frac{\nabla u_\bullet \nabla f}{f} * K_\delta\ (\bar{\varphi}(v))dv\Big).$$

We treat the second term using the key Lemma 2.4. Since $\nabla f \in C^{\alpha-1}$ and $\nabla u \in C^\alpha$, and using

$\nabla\bar{u}(\bar{\varphi}(v)) = \bar{\varphi}'(v)$, we obtain

$$\begin{aligned}
\int_s^0 \frac{\nabla u \bullet \nabla f}{f} * K_\delta\ (\bar{\varphi}(v))dv &= \int_s^0 [(\nabla \log f \bullet \nabla u) * K_\delta]\ (\bar{\varphi}(v))dv \\
&= \int_s^0 [(\nabla \log f * K_\delta) \bullet \nabla u](\bar{\varphi}(v))dv + O(s\delta^\alpha) \\
&= \int_s^0 [(\nabla(\log f * K_\delta)) \bullet \nabla \bar{u}](\bar{\varphi}(v))dv \\
&\quad + \int_s^0 [(\nabla \log f * K_\delta) \bullet [\nabla u - \nabla \bar{u}](\bar{\varphi}(v))dv + O(s\delta^\alpha) \\
&= [(\log f * K_\delta)(\bar{\varphi}(\bullet))]_s^0 + O(s\|u - \bar{u}\|_{C^1}) + O(s\delta^\alpha) \\
&= (\log f * K_\delta)(x) - (\log f * K_\delta)(\bar{\varphi}(s)) + O(s\|\bar{u} - u\|_{C^1}) + O(s\delta^\alpha).
\end{aligned}$$

In view of $S \lesssim \log(n)$ we have proved so far

$$\begin{aligned}
\mathbb{E}[\hat{f}(x)] = \int_{-S}^0 \exp\Big(- \int_s^0 \frac{1}{f} * K_\delta(\bar{\varphi}(v))dv + (\log f * K_\delta)(x) - (\log f * K_\delta)(\bar{\varphi}(s))\Big)ds \\
\times \Big(1 + O\big(\log(n)\delta^\alpha + \log(n)\|\bar{u} - u\|_{C^1}\big)\Big). \qquad (2.2)
\end{aligned}$$

Step 3. We aim to use the key identity without convolution

$$\exp\Big(- \int_s^0 \frac{1}{f}\ (\bar{\varphi}(v))dv - \log f(\bar{\varphi}(s))\Big) = \frac{d}{ds} \exp\Big(- \int_s^0 \frac{1}{f}\ (\bar{\varphi}(v))dv\Big).$$

Note that $\log(\bullet)$, $\exp(\bullet)$ and their derivatives have uniformly bounded supremum norm on compact sets in $(0, \infty)$, in particular on the range of f and $1/f$, So we have the approximation properties $(\log f * K_\delta)(x) = \log f(x) + O(\delta^\alpha)$ and $-(\log f * K_\delta)(\bar{\varphi}(s)) = \log \frac{1}{f}(\bar{\varphi}(s)) + O(\delta^\alpha)$ uniformly in x and s. We thus proceed from (2.2)

$$\begin{aligned}
\mathbb{E}[\hat{f}(x)] &= \exp\Big(\log f(x) + O(\delta^\alpha)\Big) \int_{-S}^0 \exp\Big(- \int_s^0 \frac{1}{f} * K_\delta\ (\bar{\varphi}(v))dv - (\log f * K_\delta)(\bar{\varphi}(s))\Big)ds \\
&\qquad \times \Big(1 + O\big(\log(n)\delta^\alpha + \log(n)\|\bar{u} - u\|_{C^1}\big)\Big) \\
&= f(x) \int_{-S}^0 \exp\Big(- \int_s^0 \frac{1}{f}\ (\bar{\varphi}(v))dv - \log f(\bar{\varphi}(s)) + O(\log(n)\delta^\alpha)\Big)ds \\
&\qquad \times \Big(1 + O\big(\log(n)\delta^\alpha + \log(n)\|\bar{u} - u\|_{C^1}\big)\Big) \\
&= f(x)\Big[\exp\Big(- \int_\bullet^0 \frac{1}{f}(\bar{\varphi}(v))dv\Big)\Big]_{-S}^0 \Big(1 + O\big(\log(n)\delta^\alpha + \log(n)\|\bar{u} - u\|_{C^1}\big)\Big) \\
&= f(x) + O\Big(\log(n)\delta^\alpha + \log(n)\|\bar{u} - u\|_{C^1} + \exp(-C\log(n)/(1 + r))\Big),
\end{aligned}$$

where $f \leqslant 1 + r$ and $S = C \log n$ were used. For all $C \geqslant 1 + r$ we obtain the bias estimate. □

2.3 Variance

2.3 Proposition. *Let $\bar{x}$ denote the unique minimizer of $\bar{u}$. For $x \in D$ the variance (conditional on $\bar{\varphi}$) of $\hat{f}(x)$ satisfies, for δ chosen at least as large as $\delta \gtrsim n^{-\frac{1}{2\alpha+d+3}}$,*

$$\mathrm{Var}(\hat{f}(x)) \lesssim n^{-1}\delta^{-d-4} \min\left(\log(n), \frac{\delta}{|x - \bar{x}|}\right).$$

Proof. Recall the estimator

$$\hat{f}(x) = \int_{-S}^{0} \exp\Big(- \int_{s}^{0} \Delta\hat{u}(\bar{\varphi}(v))\, dv\Big)\, ds \text{ with } \Delta\hat{u}(y) = ((\Delta K_\delta) * Y_n)(y).$$

We write $\hat{f}(x) = T(\mathbb{W})$ as a nonlinear functional of white noise, generating a cylindrical Gaussian measure on Wiener space $L^2(D)$, and plan to apply the infinite-dimensional Poincaré inequality, see e.g. [8],

$$\mathrm{Var}(\hat{f}(x)) \leqslant \mathbb{E}\left[\|DT_{\mathbb{W}}\|^2_{L^2(D)'}\right],$$

where DT_ψ denotes the Fréchet derivative of $T : L^2(D) \to \mathbb{R}$ at $\psi \in L^2(D)$.

By definition of $\hat{f}(x)$ we have for $\psi \in L^2(D)$

$$T(\psi) = \int_{-S}^{0} e_{u+n^{-1/2}\psi}(s)\, ds \text{ with } e_h(s) := \exp\Big(- \int_{s}^{0} (\Delta K_\delta) * h(\bar{\varphi}(v))\, dv\Big).$$

The Fréchet derivative of $T : L^2(D) \to \mathbb{R}$ at $\psi \in L^2(D)$ in direction $h \in L^2(D)$ is given by

$$DT_\psi[h] = - \int_{-S}^{0} e_{u+n^{-1/2}\psi}(s) \int_{s}^{0} ((\Delta K_\delta) * n^{-1/2}h)(\bar{\varphi}(v))\, dv\, ds.$$

Integration by parts yields with $E_h(v) := \int_{-S}^{v} e_h(s)ds$

$$\begin{aligned}(DT_\psi[h])^2 &= n^{-1}\Big(\int_{-S}^{0} e_{u+n^{-1/2}\psi}(s) \int_{s}^{0} ((\Delta K_\delta) * h)(\bar{\varphi}(v))\, dv\, ds\Big)^2 \\ &= n^{-1}\Big(\int_{-S}^{0} E_{u+n^{-1/2}\psi}(v)((\Delta K_\delta) * h)(\bar{\varphi}(v))\, dv\Big)^2.\end{aligned}$$

Moreover, by Cauchy-Schwarz inequality for $\|h\|_{L^2} \leqslant 1$

$$\begin{aligned}\Big(\int_{-S}^{0} E_{u+n^{-1/2}\psi}(v)((\Delta K_\delta) * h)(\bar{\varphi}(v))\, dv\Big)^2 &= \Big(\int_D \int_{-S}^{0} E_{u+n^{-1/2}\psi}(v)\Delta K_\delta(\bar{\varphi}(v) - z)h(z)\, dv\, dz\Big)^2 \\ &\leqslant \int_{\mathbb{R}^d} \Big(\int_{-S}^{0} E_{u+n^{-1/2}\psi}(v)|\Delta K_\delta(\bar{\varphi}(v) - z)|\, dv\Big)^2 dz\end{aligned}$$

holds. Therefore we have shown

$$\|DT_\psi\|^2_{L^2(D)'} \leqslant n^{-1} \int_{\mathbb{R}^d} \Big(\int_{-S}^{0} E_{u+n^{-1/2}\psi}(v)|\Delta K_\delta(\bar{\varphi}(v) - z)|\, dv\Big)^2 dz.$$

We now use $|\Delta K_\delta(y)| \lesssim \delta^{-d-2}\mathbf{1}(|y| \leqslant \delta)$ to conclude further

$$\begin{aligned}&\|DT_\psi\|^2_{L^2(D)'} \\ &\lesssim n^{-1}\delta^{-2d-4} \int_{-S}^{0}\int_{-S}^{0} E_{u+n^{-1/2}\psi}(v)E_{u+n^{-1/2}\psi}(v') \int_{\mathbb{R}^d} \mathbf{1}(|\bar{\varphi}(v) - z| \leqslant \delta,\, |\bar{\varphi}(v') - z| \leqslant \delta)\, dz\, dv\, dv' \\ &\lesssim n^{-1}\delta^{-d-4} \int_{-S}^{0} E_{u+n^{-1/2}\psi}(v') \int_{-S}^{v'} E_{u+n^{-1/2}\psi}(v)\mathbf{1}(|\bar{\varphi}(v) - \bar{\varphi}(v')| \leqslant 2\delta)\, dv\, dv'.\end{aligned}$$

From part (b) of Lemma 2.6 and the monotonicity of $E_{u+n^{-1/2}\psi}$ we deduce for $\delta \in (0, |\bar{\varphi}(v') - \bar{x}|/4)$

$$\int_{-S}^{v'} E_{u+n^{-1/2}\psi}(v)\mathbf{1}(|\bar{\varphi}(v) - \bar{\varphi}(v')| \leqslant 2\delta)\, dv \leqslant E_{u+n^{-1/2}\psi}(v') \frac{4\delta}{(1/d - \eta)|\bar{\varphi}(v') - \bar{x}|}.$$

Generally, we can bound the indicator by one so that the integral is always smaller than

$$\int_{-S}^{v'} E_{u+n^{-1/2}\psi}(v)\,dv = \int_{-S}^{v'} (v'-s)e_h(s)\,ds \leqslant SE_{u+n^{-1/2}\psi}(v').$$

Inserting both bounds for the inner integral yields with $S \sim \log n$

$$\|DT_\psi\|^2_{L^2(D)'} \lesssim n^{-1}\delta^{-d-4}\int_{-S}^{0} E_{u+n^{-1/2}\psi}(v')^2\Big(\frac{\delta}{|\bar\varphi(v')-\bar x|}\wedge\log(n)\Big)\,dv'$$

and thus for the expectation

$$\mathbb{E}\left[\|DT_{\mathbb{W}}\|^2_{L^2(D)'}\right] \lesssim n^{-1}\delta^{-d-4}\int_{-S}^{0} \mathbb{E}[E_{u+n^{-1/2}\mathbb{W}}(v')^2]\Big(\frac{\delta}{|\bar\varphi(v')-\bar x|}\wedge\log(n)\Big)\,dv'.$$

Using the exponential moment $\mathbb{E}[e^Z] = e^{\sigma^2/2}$ for $Z \sim N(0,\sigma^2)$ and $(\Delta u)*K_\delta(x) \geqslant 1-\eta$ uniformly, we obtain with a constant $C > 0$

$$\begin{aligned}
\mathbb{E}[E_{u+n^{-1/2}\mathbb{W}}(v')^2] &= \int_{-S}^{v'}\int_{-S}^{v'} \mathbb{E}\left[e^{-(\int_s^0+\int_{s'}^0)(\Delta K_\delta)*(u+n^{-1/2}\mathbb{W})(\bar\varphi(w))\,dw}\right]ds\,ds' \\
&= \int_{-S}^{v'}\int_{-S}^{v'} e^{-(\int_s^0+\int_{s'}^0)(\Delta u)*K_\delta(\bar\varphi(w))\,dw} e^{n^{-1}\|(\int_s^0+\int_{s'}^0)\Delta K_\delta(\bar\varphi(w)-\bullet)dw\|^2_{L^2}/2}ds\,ds' \\
&\leqslant \Big(\int_{-S}^{v'} e^{-\int_s^0(\Delta u)*K_\delta(\bar\varphi(w))\,dw} e^{n^{-1}\|\int_s^0\Delta K_\delta(\bar\varphi(w)-\bullet)dw\|^2_{L^2}}ds\Big)^2 \\
&\leqslant \Big(\int_{-S}^{v'} \exp\Big((1-\eta)s + Cn^{-1}\delta^{-d-4}s^2\Big)ds\Big)^2.
\end{aligned}$$

Because of $S \sim \log(n)$ and $n^{-1}\delta^{-d-4}\log(n) \to 0$, we derive further

$$\mathbb{E}[E_{u+n^{-1/2}\mathbb{W}}(v')^2] \lesssim \Big(\int_{-S}^{v'} e^{(1-2\eta)s}ds\Big)^2 \lesssim e^{2(1-2\eta)v'}.$$

Finally, $|\bar\varphi(v')-\bar x| \geqslant e^{(1/d+\eta)v'}|x-\bar x|$ from Lemma 2.6 and $\eta \leqslant 1/4$, say, imply

$$\mathbb{E}\left[\|DT_{\mathbb{W}}\|^2_{L^2(D)'}\right] \lesssim n^{-1}\delta^{-d-4}\Big(\frac{\delta}{|x-\bar x|}\wedge\log(n)\Big).$$

Poincaré's inequality gives the result. □

2.4 Mean squared error

We now complete the proof of the upper bound. With $\bar x$ denoting the minimum of $\bar u$ we have

$$\mathbb{E}[(\hat f(x)-f(x))^2] \lesssim \log(n)^2\delta^{2\alpha} + n^{-1}\delta^{-4-d}\big(\delta|x-\bar x|^{-1}\wedge\log(n)\big) + (\log n)^2\|\bar u - u_f\|^2_{C^1}.$$

The optimal (up to log-factors) choice for estimation at x is

$$\delta \sim n^{-1/(2\alpha+4+d)} \wedge (n|x-\bar x|)^{-1/(2\alpha+3+d)},$$

which is data-driven because $\bar x$ is obtained by pre-estimation. This gives the pointwise estimation rate

$$\mathbb{E}[(\hat f(x)-f(x))^2] \lesssim (\log n)^2\big(n^{-2\alpha/(2\alpha+4+d)} \wedge (n|x-\bar x|)^{-2\alpha/(2\alpha+3+d)} + \|\bar u - u_f\|^2_{C^1}\big).$$

Finally we integrate over the unit ball D to obtain the L^2-risk. We use radial coordinates around $\bar{x}$ and only use the $(n|x-\bar{x}|)^{-2\alpha/(2\alpha+3+d)}$-rate, which remains integrable, to obtain

$$\begin{aligned}\frac{1}{(\log n)^2}\int_D \mathbb{E}[(\hat{f}(x)-f(x))^2]dx &\lesssim \int_0^2 (nr)^{-2\alpha/(2\alpha+3+d)} r^{d-1}dr + \|\bar{u}-u_f\|_{C^1}^2 \\ &\sim n^{-2\alpha/(2\alpha+3+d)} + \|\bar{u}-u_f\|_{C^1}^2.\end{aligned}$$

In view of the faster pre-estimation rate from (2.1) we have thus proved with expectation over both samples

$$\boxed{\mathbb{E}[\|\hat{f}-f\|_{L^2(D)}^2] \lesssim (\log n)^2 n^{-2\alpha/(2\alpha+3+d)}.}$$

2.5 Auxiliary results

We begin with the following lemma concerning interchanging convolution and multiplication.

2.4 Lemma. *Let $f_1, f_2 : \mathbb{R}^d \to \mathbb{R}$ satisfy $f_1 \in C^{\alpha-1}$, $f_2 \in C^\alpha$ for $\alpha > 1$ and $K_\delta = \delta^{-d}K(\bullet/\delta)$ as before (with sufficiently many vanishing moments). Then we have for $x \in \mathbb{R}^d$ and $\delta \to 0$*

$$[f_1 * K_\delta](x)f_2(x) - [(f_1 f_2) * K_\delta](x) = O(\delta^\alpha).$$

The same holds for functions $f_1, f_2 : D \to \mathbb{R}$ and suitable boundary corrected kernels.

Proof. We prove the result at $x = 0$, for the non-boundary corrected case. [With boundary corrections, the proof is analogous, using Taylor expansion on $D \subseteq \mathbb{R}^d$ only, and vanishing moments of the boundary corrected kernels.] By Taylor expansion around zero we have that for $|y| \leqslant \delta$ and multi-indices $k \in \mathbb{N}_0^d$, $|k| = k_1 + \cdots + k_d$

$$\begin{aligned}f_1(y) &= \sum_{|k|\leqslant \lfloor\alpha\rfloor - 1} a_k y^k + O(\delta^{\alpha-1}) = P_1(y) + R_1(y), \\ f_2(y) &= \sum_{|k|\leqslant \lfloor\alpha\rfloor} b_k y^k + O(\delta^\alpha) = P_2(y) + R_2(y)\end{aligned}$$

with some coefficients $a_k, b_k \in \mathbb{R}$. For integer α we use the convention $\lfloor\alpha\rfloor = \alpha - 1$. Due to the vanishing moments of K_δ, we obtain

$$(f_1 * K_\delta)(0)f_2(0) = (a_0 + [R_1 * K_\delta](0))b_0 = a_0 b_0 + [R_1 * K_\delta](0)b_0,$$

as well as

$$(f_1 f_2) * K_\delta(0) = (P_1 P_2 + R_1 b_0 + R_1(P_2 - b_0) + P_1 R_2 + R_1 R_2) * K_\delta(0).$$

Since $|P_2(y) - b_0| \lesssim \delta$ holds for $|y| \leqslant \delta$, the vanishing moments of K_δ yield

$$\begin{aligned}(f_1 f_2) * K_\delta(0) &= a_0 b_0 + [R_1 * K_\delta](0)b_0 + \big[(O(\delta^\alpha) + O(\delta^\alpha) + O(\delta^{2\alpha-1})) * K_\delta\big](0) \\ &= (f_1 * K_\delta)(0)f_2(0) + O(\delta^\alpha),\end{aligned}$$

where the $O(\bullet)$-terms within the convolution denote supremum norm bounds over $|y| \leqslant \delta$. □

2.5 Lemma. *Let $\alpha > 1$. For any $\eta > 0$ there exists $r > 0$ small enough such that for all $f \in \mathcal{F}(\alpha, r)$*

$$\inf_{x\in D} \lambda_{min}(\nabla^2 u_f(x)) \geqslant 1/d - \eta/2, \quad \sup_{x\in D} \lambda_{max}(\nabla^2 u_f(x)) \leqslant 1/d + \eta/2.$$

Proof. By elliptic regularity estimates for Hölder-Zygmund spaces [12, Chapter 5] (for non-integer α, we can also use standard Schauder theory [4]) give

$$M := \sup_{f\in\mathcal{F}(\alpha,r)} \|u_f\|_{C^{\alpha+1}} < \infty.$$

Using this and $\alpha > 1$, we obtain for a suitable small $\varepsilon > 0$ the local Lipschitz estimate

$$\|u_f - u_{f'}\|_{C^{2+\varepsilon}} \leqslant L\|f - f'\|_{C^{1+\varepsilon}},$$

where L may depend on M. This is proved by a PDE perturbation argument, such as performed in [7]. The claim follows by choosing $f' = \mathbf{1}$ and by taking r small enough, in fact of order η. □

The previous lemma implies that exponential trapping lemmas on the characteristics are satisfied 'uniformly in a neighborhood' of unit diffusivity.

2.6 Lemma (Exponential convergence of characteristics)**.** *Let u be a strongly convex C^2-function on D with a unique interior minimum x_* and such that for some $\eta \in (0, 1/d)$ and every $x \in D$*

$$1/d - \eta \leqslant \lambda_{min}\big[\nabla^2 u(x)\big] \leqslant \lambda_{max}\big[\nabla^2 u(x)\big] \leqslant 1/d + \eta.$$

For fixed $x \in D$ let φ solve the gradient flow $\varphi'(t) = \nabla u(\varphi(t))$, $t \leqslant 0$, with $\varphi(0) = x$. Then:

(a) $\lim_{t\to-\infty} \varphi(t) = x^*$ holds, more precisely

$$\forall t \leqslant 0: \ e^{(1/d+\eta)t}|x - x_*| \leqslant |\varphi(t) - x_*| \leqslant e^{(1/d-\eta)t}|x - x_*|.$$

(b) For $t \leqslant s \leqslant 0$ and $\delta \in (0, |\varphi(s) - x_*|/4)$ we have the implication

$$|\varphi(s) - \varphi(t)| \leqslant 2\delta \Rightarrow s - t \leqslant \frac{4\delta}{(1/d - \eta)|\varphi(s) - x_*|}.$$

Proof. Define $y(t) := \varphi(t) - x_*$. Due to $\nabla u(x_*) = 0$ the fundamental theorem of calculus gives

$$\nabla u(\varphi(t)) = \nabla u(x_* + y(t)) - \nabla u(x_*) = \int_0^1 \nabla^2 u(x_* + \vartheta y(t))\, y(t)\, d\vartheta.$$

Taking the inner product with $y(t)$ and using the assumption on $\nabla^2 u(x)$, we have

$$(1/d - \eta)|y(t)|^2 \leqslant \langle y(t), \nabla u(\varphi(t))\rangle \leqslant (1/d + \eta)|y(t)|^2.$$

Moreover, we have $y'(t) = \varphi'(t) = \nabla u(\varphi(t))$ and therefore

$$\frac{1}{2}\frac{d}{dt}|y(t)|^2 = \langle y(t), y'(t)\rangle = \langle y(t), \nabla u(\varphi(t))\rangle.$$

Using the previous bounds, this gives the differential inequality for $F(t) := |y(t)|^2$

$$2(1/d - \eta)F(t) \leqslant F'(t) \leqslant 2(1/d + \eta)F(t).$$

Consequently, Gronwall's lemma yields

$$e^{2(1/d+\eta)t}F(0) \leqslant F(t) \leqslant e^{2(1/d-\eta)t}F(0), \quad t \leqslant 0,$$

which proves (a).

For (b) we obtain from above by the reverse triangle inequality

$$|\varphi(s) - \varphi(t)| = |y(s) - y(t)| \geqslant |y(s)| - |y(t)| \geqslant |y(s)|\big(1 - e^{(1/d-\eta)(t-s)}\big).$$

Therefore, $|\varphi(s) - \varphi(t)| \leqslant 2\delta$ implies $1 - 2\delta/|y(s)| \leqslant e^{(1/d-\eta)(t-s)}$ and hence

$$s - t \leqslant \frac{1}{1/d - \eta}\log\left(\frac{|y(s)|}{|y(s)| - 2\delta}\right) \leqslant \frac{1}{1/d - \eta}\log\left(1 + \frac{4\delta}{|y(s)|}\right) \leqslant \frac{4\delta}{(1/d - \eta)|y(s)|},$$

using $2\delta \leqslant |y(s)|/2$. □

2.6 Lower bound

We prove the lower bound using multiple hypothesis testing techniques, see, e.g., [13].

Step 1: Construction of alternatives. We employ spherical coordinates and write $x \in D \setminus \{0\}$ as (r, ω) with $r = |x| \in (0, 1]$ and $\omega = x/|x| \in S_{d-1}$ (the radius r should not be confused with the bound r on the Hölder norm). The Laplace operator can then be expressed as

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{d-1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\Delta_{S_{d-1}}$$

with $\Delta_{S_{d-1}}$ denoting the Laplace-Beltrami operator on the sphere.

Let $\omega_1, \dots, \omega_m \in S_{d-1}$ be a maximal (2δ)-packing of S_{d-1} with respect to Euclidean distance in $\mathbb{R}^d$. Then $m \sim \delta^{-(d-1)}$ holds. For some $c > 0$ we define

$$\psi_i(r, \omega) = c\delta^{\alpha+2} a(r) b((\omega - \omega_i)/\delta), \quad i = 1, \dots, m,$$

with a smooth function $a : [0, 1] \to \mathbb{R}$, satisfying $a \geqslant 0$, $\operatorname{supp}(a) \subseteq [1/4, 3/4]$, $\|a\|_{L^2} = 1$, and a smooth function $b : \mathbb{R}^d \to \mathbb{R}$, satisfying $\operatorname{supp}(b) \subseteq D$, $\|b\|_{L^2} = 1$. Note that the supports of the ψ_i are mutually disjoint. For any sign vector $\varepsilon \in \{-1, +1\}^m$ consider

$$u_\varepsilon(r, \omega) = u_0(r, \omega) + \sum_{i=1}^m \varepsilon_i \psi_i(r, \omega) \text{ with } u_0(r, \omega) = \frac{r^2 - 1}{2d}.$$

Denoting by P_ε the observational law of Y_n under u_ε and by P_0 the one under u_0, we have

$$\mathrm{KL}(P_\varepsilon \,|\, P_0) \leqslant n\|u_\varepsilon - u_0\|_{L^2}^2 = c^2 n \sum_{i=1}^m \delta^{2\alpha+4} \int_0^1 a(r)^2 r^{d-1} dr \int_{S_{d-1}} b((\omega - \omega_i)/\delta)^2 \, d\omega \sim nm\delta^{2\alpha+4}\delta^{d-1}.$$

For a suitable choice of $\delta \sim n^{-1/(2\alpha+3+d)}$ this is bounded by a small constant times m and the classical lower bound technique in L^2, see [13], yields the asserted lower bound, provided that the diffusivities f_ε associated to u_ε satisfy

(a) $f_\varepsilon \in \mathcal{F}(\alpha, r)$;

(b) $\|f_\varepsilon - f_{\varepsilon'}\|_{L^2}^2 \gtrsim \delta^{2\alpha} m^{-1} \sum_{i=1}^m \mathbf{1}(\varepsilon_i \neq \varepsilon_i')$.

Step 2: Signal separation. We first assume that the diffusion coefficient is obtained by integrating along the radial gradient flow $\varphi_0(t) = xe^{t/d}$ instead of φ_ε with $\varphi_\varepsilon'(t) = \Delta u_\varepsilon(\varphi_\varepsilon(t))$. Consider the associated 'approximate' inversion formula

$$\tilde{f}_\varepsilon(x) = \int_{-\infty}^0 \exp\Big(- \int_s^0 \Delta u_\varepsilon(\varphi_0(v))\, dv \Big)\, ds.$$

By a change of variables in v and s and inserting $\Delta u_0 = 1$, this can be written as

$$\begin{aligned}\tilde{f}_\varepsilon(x) &= \int_0^1 \exp\Big(- d\int_{s'}^1 \Delta u_\varepsilon(v'x) \frac{dv'}{v'} \Big) \frac{ds'}{s'} \\ &= \int_0^1 d(s')^{d-1} \exp\Big(- d\sum_{i=1}^m \varepsilon_i \int_{s'}^1 \frac{\Delta\psi_i(v'x)}{v'}\, dv' \Big)\, ds'.\end{aligned}$$

Concerning the Hölder regularity of $\tilde{f}_\varepsilon$, note that $\|\Delta\psi_i\|_{C^\alpha} \lesssim 1$ holds due to the $\delta^{\alpha+2}$-scaling. Moreover, for each $x = (r, \omega) \in D$ there is at most one ω_i with $|\omega_i - \omega| < \delta$ such that at most one term in the sum inside the exponential is non-zero. Due to $\Delta\psi_i(v'x) = 0$ for $v'|x| \leqslant 1/4$ division by v' does not harm. We conclude $\|\tilde{f}_\varepsilon - \mathbf{1}\|_{C^\alpha} \lesssim c\|\Delta\psi_i\|_{C^\alpha} \lesssim c$ and for each $r > 0$ we can choose $c > 0$ sufficiently small to guarantee $\|\tilde{f}_\varepsilon - \mathbf{1}\|_{C^\alpha} \leqslant r/2$.

We bound $\|\tilde{f}_\varepsilon - \tilde{f}_{\varepsilon'}\|_{L^2}^2$ from below. Consider $x = (r, \omega)$ with $|\omega - \omega_i| \leqslant \delta$ for some (necessarily unique) ω_i. Whenever $\varepsilon_i \neq \varepsilon_i'$,

$$|\tilde{f}_\varepsilon(x) - \tilde{f}_{\varepsilon'}(x)| = \Big| \int_0^1 \Big(\exp\Big(- d \int_{sr}^r \frac{\Delta \psi_i(v\omega)}{v}\, dv \Big) - \exp\Big(- d \int_{sr}^r \frac{\Delta \psi_i(v\omega)}{v}\, dv \Big) \Big) ds^{d-1} ds \Big|$$

holds. We evaluate, using the quadratic scaling of the Laplace-Beltrami operator in the spherical coordinates,

$$\begin{aligned} \int_s^1 \frac{\Delta \psi_i(vx)}{v}\, dv = c\delta^{\alpha+2} \Big(\int_{sr}^r \frac{a''(v) + (d-1) r^{-1} a'(v)}{v} dv \Big) b((\omega - \omega_i)/\delta) \\ + c\delta^\alpha \Big(\int_{sr}^r \frac{a(v)}{r^2 v} dv \Big) \Delta_{\delta^{-1} S_{d-1}} b((\omega - \omega_i)/\delta) \end{aligned}$$

We thus obtain uniformly over x with $|\omega - \omega_i| \leqslant \delta$ as $\delta \to 0$

$$\frac{|f_\varepsilon(x) - f_{\varepsilon'}(x)|}{\delta^\alpha} \to c \int_0^1 \Big(2d \int_{sr}^r \frac{a(v)}{r^2 v} dv \Big) ds^{d-1} ds |\Delta_{\delta^{-1} S_{d-1}} b((\omega - \omega_i)/\delta)|,$$

noting that a is a non-negative function. For $r \geqslant 1/2$ and $|\omega - \omega_i| \leqslant \delta/2$, say, the right-hand side is uniformly bounded away from zero. This shows

$$\delta^{-2\alpha} \|f_\varepsilon - f_{\varepsilon'}\|_{L^2(D)}^2 \sim \int_{1/2}^1 \int_{S_{d-1}} \sum_{i : \varepsilon_i \neq \varepsilon_i'} \mathbf{1}(|\omega - \omega_i| \leqslant \delta/2) r^{d-1} dr d\omega \sim \sum_{i : \varepsilon_i \neq \varepsilon_i'} \delta^{d-1} \sim m^{-1} \sum_{i=1}^m \mathbf{1}(\varepsilon_i \neq \varepsilon_i'),$$

as required.

Step 3. Non-radial characteristics. We have by definition

$$\sup_\varepsilon \|u_\varepsilon\|_{C^{\alpha+2}} \lesssim 1, \quad \|u_\varepsilon - u_0\|_{C^1} = \Big\| \sum_{i=1}^m \varepsilon_i \psi_i(r, \omega) \Big\|_{C^1} \lesssim \delta^{\alpha+1},$$

since the ψ_i have disjoint support. By choosing the constant c in ψ_i small enough, we may assume that the u_ε are all strictly convex with convexity parameter at least $1 - \eta$ for any $\eta \in (0, 1)$.

Given $x \in D$ let φ_ε be the characteristic given by $\varphi_\varepsilon'(t) = \nabla u_\varepsilon(\varphi_\varepsilon(t)), t \leqslant 0$, and φ_0 correspondingly for u_0. By Lemma 2.6 there exists some t_0 such that for all $x \in D$, all ε and all $t \leqslant -t_0$, $|u_\varepsilon(\varphi_\varepsilon(t))| < 1/4$. For all $z \in B_{1/4}(0)$ (open ball) we have that $\nabla u_\varepsilon(z) = \nabla u_0(z) = x/d$. Moreover, note that the gradient flow of u_0 is contractive, i.e. if φ_1, φ_2 are characteristics of ∇u_0 starting at $y_1, y_2 \in D$, respectively, then $|\varphi_1(s) - \varphi_2(s)| \leqslant |y_1 - y_2|$. By a Grönwall argument, we have that up until time t_0, for some constant C, we have the estimate

$$\sup_{v \in [-t_0, 0]} \|\varphi_\varepsilon(v) - \varphi_0(v)\| \leqslant C \|\nabla u_\varepsilon - \nabla u_0\|_{L^\infty} \leqslant C \|u_\varepsilon - u_0\|_{C^1} \lesssim \delta^{\alpha+1}.$$

For all times $t \leqslant -t_0$ the same holds by contractivity such that we have the *global* in time bound

$$\sup_{v \in (-\infty, 0]} \|\varphi_\varepsilon(v) - \varphi_0(v)\| \lesssim \delta^{\alpha+1}.$$

Let us denote by $\tilde{f}_\varepsilon(x)$ the previous reconstruction under the radial characteristic φ_0. From $\Delta u_\varepsilon \geqslant 1 - \eta$ and $x \mapsto \exp(-x)$ globally Lipschitz for $x \geqslant 0$ we deduce

$$\begin{aligned} |\tilde{f}_\varepsilon(x) - f_\varepsilon(x)| &= \Big| \int_{-\infty}^0 \exp\Big(- \int_s^0 \Delta u_\varepsilon(\varphi_0(v)) dv \Big) ds - \int_{-\infty}^0 \exp\Big(- \int_s^0 \Delta u_\varepsilon(\varphi_\varepsilon(v)) dv \Big) ds \Big| \\ &\leqslant \int_{-\infty}^0 e^{s(1-\eta)} \int_s^0 \|u_\varepsilon\|_{C^{2+\alpha}} |\varphi_0(v) - \varphi_\varepsilon(v)|\, dv\, ds \\ &\lesssim \sup_{v \in (-\infty, 0]} \|\varphi_\varepsilon(v) - \varphi_0(v)\| \int_{-\infty}^0 s e^{s(1-\eta)} ds \lesssim \delta^{\alpha+1}. \end{aligned}$$

This holds uniformly for all x and is of smaller order than the δ^α separation shown previously.

Step 4: Hölder constraint. Finally, let us prove that $f_\varepsilon \in C^\alpha$ and restrict to integer smoothness $\alpha \in \mathbb{N}$ to remain concise. We write

$$f_\varepsilon = \int_{-\infty}^0 \exp\Big(\int_s^0 \Delta u_\varepsilon \circ \Phi_v dv\Big) ds,$$

where $\Phi_v : D \to D$ is the time-v flow map of ∇u_ε. We will show that $\sup_{v\leqslant 0} \|\Phi_v\|_{C^\alpha} \leqslant C$. This will imply $f_\varepsilon \in C^\alpha$ by the chain rule: when D^m denotes the tensor of m-fold derivatives, there is for any multi-index β, $|\beta| \leqslant \alpha$, a multivariate polynomial P_β of order at most α with

$$\begin{aligned}\|\partial^\beta f_\varepsilon\|_\infty &= \Big\| \int_{-\infty}^0 P_\beta(\Delta u_\varepsilon \circ \Phi_v, D[\Delta u_\varepsilon \circ \Phi_v], \dots, D^\alpha[\Delta u_\varepsilon \circ \Phi_v]) \exp\Big(\int_s^0 \Delta u_\varepsilon \circ \Phi_v \, dv\Big) \, ds\Big\|_\infty \\ &\lesssim \int_{-\infty}^0 e^{(1-\eta)s} ds, \end{aligned} \tag{2.3}$$

since $\sup_{\varepsilon,v}(\|\Delta u_\varepsilon\|_{C^\alpha} + \|\Phi_v\|_{C^\alpha}) = O(1)$ and $\Delta u_\varepsilon(x) \geqslant 1-\eta$ uniformly over x for any $\eta > 0$ when choosing $c = c(\eta) > 0$ sufficiently small in the definition of u_ε.

We prove $\sup_{v\leqslant 0} \|\Phi_v\|_{C^\alpha} \leqslant C$ by induction over $\alpha \in \mathbb{N}$. Start with $\alpha = 1$ and note that

$$\partial_t D\varphi_\varepsilon = D\dot\varphi_\varepsilon = D(Du_\varepsilon \circ \varphi_\varepsilon) = (D^2 u_\varepsilon \circ \varphi_\varepsilon) D\varphi_\varepsilon, \quad t \leqslant 0,$$

applying the derivative tensors D, D^2 to the solution u_ε and the flow φ_ε appropriately, with terminal condition $D\dot\varphi_\varepsilon(0) = \mathrm{Id}$. Thus, with the HS-norm $\|\bullet\|_{HS}$,

$$\partial_t \|D\varphi_\varepsilon\|_{HS}^2 = 2\langle D\varphi_\varepsilon, (D^2 u_\varepsilon \circ \varphi_\varepsilon) D\varphi_\varepsilon\rangle_{HS} \geqslant 2(1/d - \eta)\|D\varphi_\varepsilon\|_{HS}^2,$$

where we used that $(D^2 u_\varepsilon \circ \varphi) \succeq (1/d - \eta)\,\mathrm{Id}$ uniformly. By Gronwall's inequality in backward time and using that $\|D\varphi_\varepsilon(0)\|_{HS}^2 = d$, we obtain

$$\|D\varphi_\varepsilon(t)\|_{HS}^2 \leqslant d e^{2(1/d-\eta)t}, \quad t \leqslant 0,$$

which proves $\|\Phi_t\|_{C^1} \leqslant C$ uniformly over t.

For the induction step $(k-1) \mapsto k$, $k = 1, \dots, \alpha+1$, assume that there exists some constant $C_{k-1} < \infty$ with $\|\Phi_t\|_{C^{k-1}} \leqslant C_{k-1}$. We use the chain rule repeatedly to assert

$$\begin{aligned} D^k\dot\varphi_\varepsilon &= D^k(Du_\varepsilon \circ \varphi_\varepsilon) \\ &= (D^{k+1}u_\varepsilon \circ \varphi_\varepsilon)(D\varphi_\varepsilon)^{\otimes k} + (D^k u_\varepsilon \circ \varphi_\varepsilon) \sum_{l=0}^{k-2} a_l (D\varphi_\varepsilon)^{\otimes l} \otimes (D^2\varphi_\varepsilon) \otimes (D\varphi_\varepsilon)^{\otimes(k-2-l)} \\ &\quad + \dots + (D^2 u_\varepsilon \circ \varphi_\varepsilon) D^k \varphi_\varepsilon \end{aligned} \tag{2.4}$$

with real coefficients a_l and further terms involving $D^l u_\varepsilon$, $l = 3, \dots, k-1$, and derivatives of φ_ε up to order $k-1$. Only the final term involves $D^k\varphi_\varepsilon$. Since all third partial derivatives of u_0 vanish, and $\|u_\varepsilon - u_0\|_{C^{\alpha+2}} \leqslant \eta$ can be achieved by a small choice of $c = c(\eta)$, the induction hypothesis yields

$$\partial_t D^k\varphi_\varepsilon = D^k\dot\varphi_\varepsilon = -\mathcal{O}(\eta) + (D^2 \circ u_\varepsilon) D^k\varphi_\varepsilon$$

Taking, as before for matrices, a scalar product between tensors and denoting the constant involved in $\mathcal{O}(\eta)$ under the norm by C_k, we obtain as for the case $k = 1$ that with another constant C_k'

$$\partial_t \|D^k\varphi_\varepsilon\|^2 \geqslant 2(1/d-\eta)\|D^k\varphi_\varepsilon\|^2 - C_k\eta\|D^k\varphi_\varepsilon\| \geqslant (1/d-\eta)\|D^k\varphi_\varepsilon\|^2 - C_k'\eta^2.$$

Gronwall's inequality yields

$$\|D^k\varphi_\varepsilon\|^2 \leqslant \int_t^0 e^{(1/d-\eta)(t-s)} C_k'\eta^2 ds \leqslant \frac{C_k'\eta^2}{1/d-\eta},$$

which again remains bounded. By induction, we conclude $\sup_{\varepsilon,v} \|\Phi_v\|_{C^{\alpha+1}} \leqslant C$.

Finally we show that a sufficiently small choice of $c > 0$ can ensure $\|f_\varepsilon - \mathbf{1}\|_{C^\alpha} \leqslant r$. Above we have seen that we can assure $\|\tilde{f}_\varepsilon - \mathbf{1}\|_{C^\alpha} \leqslant r/2$ for the approximation along the radial flow and it remains to show $\|f_\varepsilon - \tilde{f}_\varepsilon\|_{C^\alpha} \leqslant r/2$. This can be seen from the representation (2.3) as well as (2.4). Indeed $\partial^\beta \tilde{f}_\varepsilon$ can be represented in the same form using the flow $\Phi_{0,v}$ associated to the radial characteristics $\varphi(\bullet)$,

$$\partial^\beta \tilde{f}_\varepsilon = \int_{-\infty}^0 P_\beta(\Delta u_\varepsilon \circ \Phi_{0,v}, D[\Delta u_\varepsilon \circ \Phi_v], \dots, D^\alpha[\Delta u_\varepsilon \circ \Phi_{0,v}]) \exp\Big(\int_s^0 \Delta u_\varepsilon \circ \Phi_{0,v} dv\Big) ds,$$

and $D^k \dot{\varphi}$ satisfies (2.4) with u_0, φ in place of $u_\varepsilon, \varphi_\varepsilon$. Since $\|u_\varepsilon - u_0\|_{C^{\alpha+2}} \leqslant \eta$ we can ensure $\|\Phi_v - \Phi_{0,v}\|_{C^\alpha} \lesssim \eta$ for any fixed η. This finally gives $\|\partial_\beta(f_\varepsilon - \tilde{f}_\varepsilon)\|_\infty \lesssim \eta$ and thus $\|f_\varepsilon - \tilde{f}_\varepsilon\|_{C^\alpha} \leqslant r/2$, by choosing a sufficently small $\eta > 0$ and $c = c(\eta) > 0$.

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